Supplementary material: Text properties and children's reading pace in a multilingual storytelling app

S1. The theory object

The constructs, causal claims, auxiliary assumptions, rival accounts and predictions behind this protocol are held in a single machine-readable file, `theory/intract.theory.yaml`, which the theoryforge package reads and checks (Bernabeu, 2026b). Figure 2 of the main text is emitted from that file, so the diagram cannot disagree with the hypotheses. Running `theory/build_theory.R` reproduces everything in this section.

The object registers twelve constructs, each with a definition, its intended measurement and its boundary conditions. Eighteen propositions link them, each stating its mechanism as well as its direction. Fifteen of the eighteen are directed edges of the causal graph, and the other three are moderation claims. Six auxiliary assumptions are recorded as entries in their own right, five of them corresponding to questions put to the data controller and listed in Table 2 of the main text. Six rival accounts are registered, and seven of the ten predictions are marked diagnostic against one of them.

The rigour checklist scores the object at 90.9 out of 100 with no blocking failure, and every item passes. On falsifiability and precision, at least one prediction forbids an observation, all ten predictions are interval predictions, and their mean severity clears the threshold. On construct quality, every construct carries a definition, a measurement and boundary conditions, no pair exceeds the lexical redundancy screen (largest observed similarity 0.14), and the theory's scope is stated. On structure, every proposition states a mechanism, at least one relation is causal, auxiliary assumptions are fewer than propositions with none added defensively in response to a prediction, and every prediction is reachable from a proposition in the graph. Finally, at least one prediction discriminates between registered alternatives, and a formal-model reference points at the design specification used for the precision analysis.

Because the causal graph is acyclic it yields testable implications, thirty of them, of which three are tested in the fourth outcome-neutral quality check of the main text. The most consequential is that the three text properties should be mutually independent once the reading level and the generation route are conditioned on. A first version of the object, lacking the text-provenance construct, implied that they were independent unconditionally, which is false of any real corpus: a text's target reading level shifts all three property families at once. Adding text provenance as a common cause of all three, and giving it an incoming path from the reader's own decoding automaticity, both repaired that implication and derived the adjustment set the analysis uses.

S2. Studies from the accompanying systematic review

The Introduction summarises the evidence that motivates the feature set and the moderators. Supplementary Table 1 lists the studies closest to this design in method, meaning a broad set of text features entered into a

statistical or machine-learning model to predict a behavioural or comprehension outcome in children.

Supplementary Table 1
Studies modelling children's reading computationally from text features

Study	Sample and design	Method and key findings
Pickren et al. (2022)	Children (*N* = 158, ages 8;8 to 11;2); multilevel logistic models	Word frequency, pronoun density, concreteness and deep cohesion each predicted passage comprehension after decoding and language comprehension were controlled, and lexical arousal predicted it beyond those.
Aparicio et al. (2022)	Children (*N* = 1,229, grades 2 to 9); multivariate regression	Text and sentence characteristics predicted comprehension, but the direction and size of each effect varied by question type.
Ludewig et al. (2023)	Children (*N* = 924, grade 4); non-linear regression	Estimated lexical text coverage predicted comprehension non-linearly, rising more steeply above roughly 56 per cent coverage.
Nie et al. (2022)	Elementary school students; feature-based prediction model	Sentence complexity, lexical surprisal and rare-word features improved prediction of item-level comprehension scores over a baseline.
Wasito Kirana et al. (2025)	Bilingual children (*N* = 44, 22 with language impairment); eye-tracking	Syntactic complexity impaired comprehension accuracy and slowed recall, disproportionately for children with language impairment.

Supplementary Table 2 lists the digital and computerised reading studies closest to this design in modality and measurement. Most come from the accompanying systematic review, with one directly comparable app-log precedent retained from outside it (Sortkær et al., 2021).

Supplementary Table 2
Digital and app-based reading studies informing the present design

Study	Platform and sample	Data, focus and key findings
Sortkær et al. (2021)	Educational app (BookBites); 10,903 students in grades 0 to 9 contributing 215,521 reading sessions, with test-score comparisons for users in grades 3, 5 and 7	App reading time against standardised test scores. Reading speed correlated with ability (*r* = .39 to .48); models predicted 16 to 18 per cent of variance in progress.
Gissel and Andersen (2021)	Digital text-to-speech tool; *N* = 1,013 children, 47 classrooms, grade 2	Cluster-randomised trial against an active phonics control. No difference in decoding, but the digital tool improved comprehension, which is the outcome most sensitive to text-level scaffolding.
Topping (2018)	Computerised book-reading assessment; *N* = 852,295 students, 3,243 schools	Quiz-based comprehension checks linked to a computerised adaptive test. Higher quiz engagement went with higher achievement, but the engagement measure itself was of limited reliability.

Study	Platform and sample	Data, focus and key findings
Bulut et al. (2023)	Online reading assessment; multi-country sample of fourth graders (ePIRLS 2016)	Explanatory item response modelling of item format, text complexity and cognitive complexity. Item properties and student characteristics interacted to predict online comprehension.
Keelor et al. (2023)	Text-to-speech study; children aged 8 to 12 with reading and language difficulties	Within-subject comparison of silent reading, listening only and text-to-speech. Text-to-speech improved comprehension over silent reading, with the benefit concentrated in children with dyslexia.
Tüchler and Cain (2026)	Digital against print medium; N = 120 children, grades 4 to 6	Generalised linear mixed models of medium and text structure. Comprehension was lower in the digital condition after word recognition and vocabulary were controlled, a screen-inferiority effect relevant to interpreting any digital-only dataset.
Beigman Klebanov and Hoang (2026)	App-embedded stealth assessment (Relay Reader)	Machine-learning models using text and oral-reading features predicted passage comprehension with over 80 per cent precision, showing that unobtrusive in-app comprehension prediction is feasible.

S3. Language resources for the text features

A multilingual design faces a measurement problem that a monolingual one does not. Concreteness taken from a 40,000-item English rating study, a 350,000-lemma German regression, a 1,400-item Spanish rating study and a 77-language embedding prediction are four different measurements wearing one label. If a resource is chosen language by language on the grounds of local quality, then the method of measurement is perfectly confounded with the language, and standardising within language removes the difference in location and scale but not the difference in construct. There are two clean responses: use one uniformly produced resource everywhere, accepting lower absolute quality for higher comparability, or use native norms and treat every cross-language comparison as descriptive. We take the first, because language enters the confirmatory models as a grouping factor with varying slopes, and those slopes have to mean the same thing in every language. The native-norm sensitivity analysis and the per-language correlation between the two measures are specified in the main text. That correlation is the quantity that tells a reader how much the choice mattered.

Two consequences of that rule revise what an earlier version of this protocol assumed. The `wordfreq` package is not used as the primary frequency source: its data are a frozen snapshot of usage to about 2021, its maintainer discontinued it in September 2024 to avoid ingesting post-2022 web text contaminated by generated language, and its wordlists for Korean and Greek are built from fewer sources than the rest. The non-English editions of the NRC Valence, Arousal and Dominance lexicon are not used at all: they are machine translations of the English norms made in 2022, not ratings collected in the target languages, and their licence is non-commercial.

A subtitle corpus indexes adult exposure, and the register mismatch for developing readers is a limitation that no available uniform resource removes. A recent demonstration that frequency estimates from a large-language-model corpus prompted for child-directed text fit young German readers better than child-book norms did (Schepens et al., 2025) sharpens that point without changing the choice: a generated corpus is neither frozen nor documented in the way a subtitle pipeline is, and the same study notes that such corpora show atypical linguistic patterns. We therefore keep subs2vec as the primary resource. Wherever a native

child-register norm exists for a language, we will report the per-language correlation between the primary frequency estimate and that norm, so that the cost of the adult register is visible in the language where it can be measured.

Supplementary Table 3 gives the primary resource for each feature family, the sensitivity resource, and the coverage across the ten languages the structural sample showed. Those ten languages are Dutch, English, French, German, Greek, Italian, Korean, Polish, Portuguese and Spanish, and every per-language claim in the table refers to that list. Where the export turns out to carry more languages, the same rules are applied to them and the table is extended to cover them.

Supplementary Table 3

Resources for the text features, the uniform primary and the native sensitivity

Feature family	Primary, uniform across languages	Sensitivity, native where it exists	Coverage of the ten sampled languages
Word frequency	subs2vec, one OpenSubtitles pipeline applied identically to 55 languages (van Paridon & Thompson, 2021)	The SUBTLEX family, native per language	All ten for the primary. Native norms exist for nine, eight of them peer-reviewed and the Italian one a conference release; Korean has no subtitle-based norm.
Age of acquisition	The cross-linguistic norms of Łuniewska and colleagues, an identical 299-word item set rated on an identical scale in 32 languages (Łuniewska et al., 2016; Łuniewska et al., 2019)	Broad native norm sets, which exist at scale for English, Dutch and Polish only	Seven of ten for the anchor set. French, Portuguese and Korean appear in neither paper. Because 299 words is thin for a text-level mean, age of acquisition is not an anchor for the lexical component and contributes only where a broad native set exists.
Surprisal	Goldfish, an identical 125-million-parameter architecture trained on byte-premium-scaled equal-sized data for each language (Chang et al., 2026)	mGPT, as a single multilingual model rather than one model per language	All ten at the 1 GB tier. This is the one feature for which a comparable multilingual resource exists by construction.
Word length, lexical density, lexical diversity	Computed from the text directly, so no norm is involved	Not applicable	All ten.
Concreteness	Native human ratings only; no value is predicted or projected	Not applicable, since the native norm is the only acceptable measure	Eight of ten, at between roughly 1,100 and 40,000 items (Brysbaert et al., 2014). Greek and Korean have only small rated sets, of 348 and 450 words, embedded in affective norms, which cannot meet the 80 per cent coverage rule, so concreteness is not computed for either language and neither contributes to any concreteness analysis.

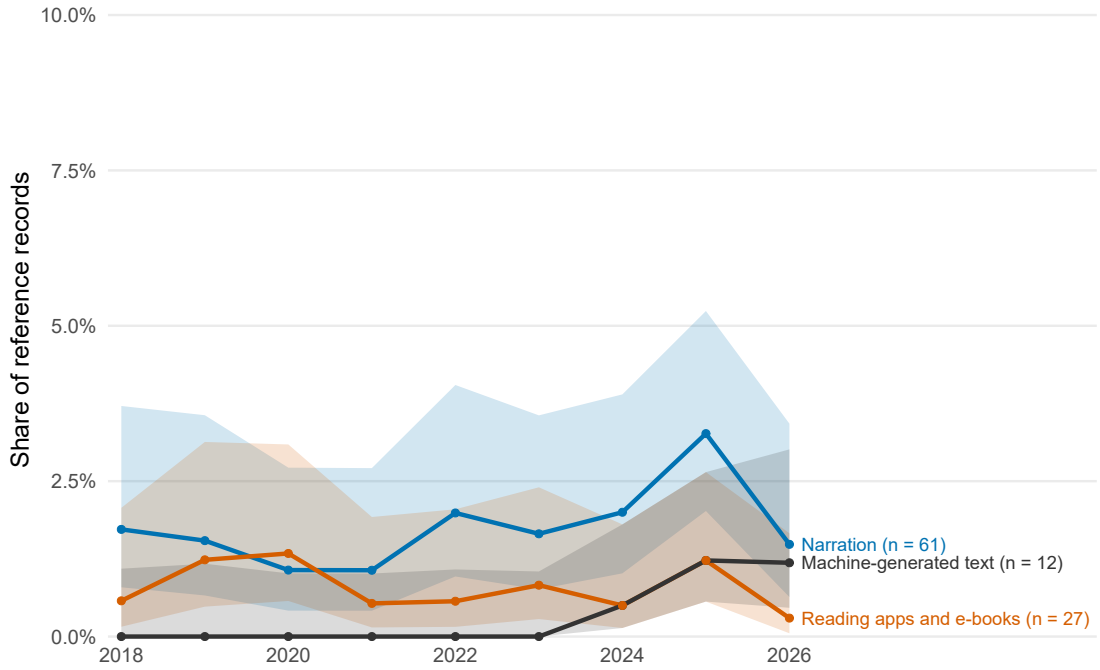
Feature family	Primary, uniform across languages	Sensitivity, native where it exists	Coverage of the ten sampled languages
Dependency distance and clause structure	Stanza (Qi et al., 2020), whose labelled attachment scores on its performance page, for single-treebank models at version 1.5.1 on Universal Dependencies 2.12, range across these ten languages from 81.8 on German GSD to 93.6 on German HDT, with Korean the lowest language at 85.5	UDPipe 2, whose current models track a later Universal Dependencies release and, on their own later test sets, score higher on nine of the ten matched treebanks and level on French GSD	All ten in both. Both parsers will be run, and the per-language correlation between the two dependency-distance estimates will be reported with the feature-pipeline validation. A language in which they disagree materially is flagged and its grammatical component is not used.
Adjacent-sentence similarity	BGE-M3, peer-reviewed, over 100 languages, and able to take a whole passage as input (Chen et al., 2024)	multilingual-E5-large-instruct, identified by checkpoint, since no DOI is registered	All ten in both. LaBSE is not used: it is optimised for retrieving translation pairs rather than for semantic similarity within a language (Feng et al., 2022).
Referential overlap, connective density	Computed from the parse and from language-specific connective inventories compiled for this study and deposited with it	Not applicable	All ten, with the inventory for each language checked by a native speaker before use.
Valence and arousal	MEmoLon, induced for 91 languages from a single procedure, at about two million entries for each of these ten (Buechel et al., 2020)	Native rated norms, which exist for all ten, though the Korean sets hold 267 to 450 words and the Greek sets 348 to 1,034	All ten for the primary. The Korean and Greek rated sets are small, so the sensitivity analysis in those languages is subject to the 80 per cent coverage rule.

Korean has the fewest resources of the ten languages, and we record the consequences here in advance. Of the seven norm-based feature families, Korean lacks a subtitle frequency norm and any coverage in the cross-linguistic age-of-acquisition set, and its rated concreteness and valence and arousal norms are sets of 267 to 450 words published in Korean-language journals, too small to meet the 80 per cent coverage rule. What it does have is a uniform frequency estimate, a Goldfish model, both parsers, induced affect and a lexical decision megastudy of 30,930 words (Yi et al., 2017). Korean therefore contributes to the lexical and grammatical components and to the affective predictors through the uniform resources, and contributes to no analysis that requires a native norm. A search of Crossref is a poor instrument for Korean-language psycholinguistics, so before the features are frozen a Korean-reading collaborator will search the Korea Citation Index and RISS directly, and any larger norm set found there will be added to the sensitivity analysis with the addition recorded as a deviation from this protocol. Greek is the next weakest. Its rated concreteness and affect sets are likewise small, 348 words in one and 1,034 translated items in another, and it has no CHILDES collection. French and Portuguese are otherwise well resourced but absent from both age-of-acquisition papers, so age of acquisition contributes for them only through the broad native sets, where those exist.

Every tool will be recorded by name, version and, where the tool is a model, by the exact checkpoint. Several of the tools named here have no registered DOI, which is normal for research software distributed through model repositories. Those are identified by checkpoint instead, and the identifiers are deposited with the feature-extraction code.

S4. The literature sizing

The Scopus count reported in the Introduction is reproduced verbatim here: the queries behind it are recorded in `literature/scopus_search_record.md`, together with the date of the search and the package version. Only count queries were issued, one request per cell, so nothing was downloaded and no record set was assembled. Supplementary Figure 1 shows how the surrounding topics have moved over the window.

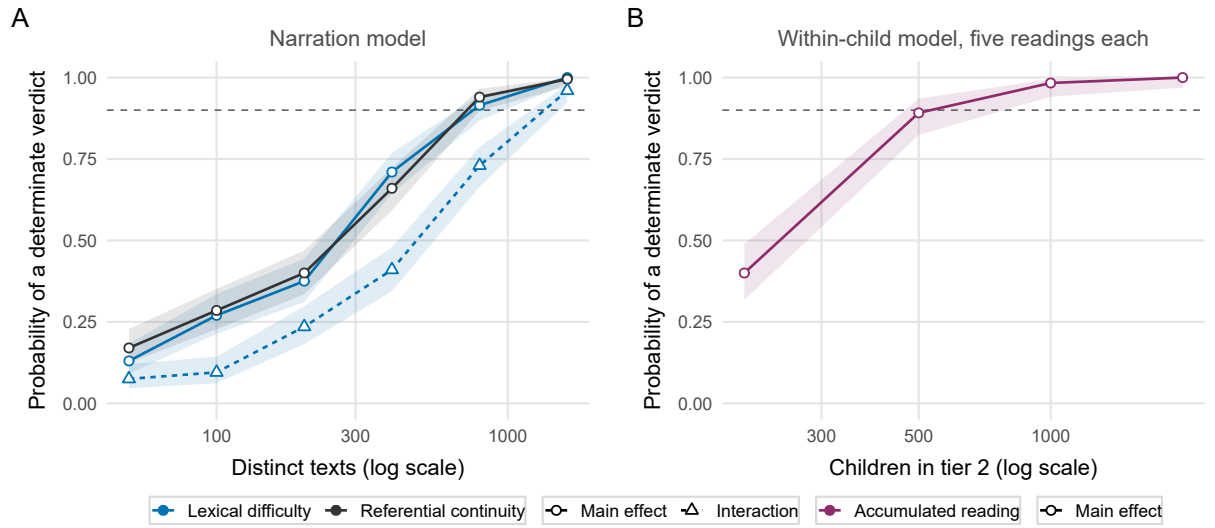


Supplementary Figure 1
Publication trend, 2018 to 2026, for children's reading comprehension in combination with narration, with reading apps and e-books, and with machine-generated text, as a percentage of the reference set. Machine-generated text does not appear in this literature at all before 2024.

S5. The design analysis in full

The precision analysis simulated data from the portable design specifications deposited in `design_analysis/`, and `design_analysis/run_design_analysis.R` reproduces it (Bernabeu, 2026a). The simulated tier-1 model contained the three components, the age band, the two age-band interactions and a text intercept, and was fitted by maximum likelihood with lme4. The narration model contained the lexical and continuity components, the narration flag and their interactions with it, and a text intercept. The tier-2 model added a child intercept and a single accumulated-reading trend. None of the simulated models carried the reading-level adjustment, the covariates, the language, country and app-version intercepts, the language-varying slopes, the distributional sub-model or the review-derived priors that the registered models carry, and the main text states what follows from that. The pooled grid is in `design_analysis/precision_sweep.csv`. The per-sweep files under `design_analysis/sweeps/` add, for each grid point, the Monte Carlo standard error on each decision probability, the number of fits attempted and the counts of converged, singular and warning fits. The texts and narration sweeps used 200 replicates per grid point, and the readings, null and tier-2 sweeps used 120. Supplementary Figure 2 shows the narration and within-child curves that the main text summarises.

Nature Human Behaviour names 0.95 as the a priori power floor for frequentist plans. We register 0.90 for this



Supplementary Figure 2
Precision for the narration contrast (a) and the within-child accumulated-reading term (b). Points are the probability of a determinate verdict for an effect of 0.10 standard deviations, over 200 simulation replicates per grid point in a and 120 in b, with twelve readings per text in a and five readings per child in b; shaded bands are 95 per cent Wilson score intervals on each probability, and the dashed line marks the 0.90 target. The narration curve is drawn for a balanced contrast and has to be read against the design effect for the observed narration prevalence, which is given in the main text.

estimation design, for the reasons given in the main text, and `design_analysis/solve_at_095.R` re-solves the same archived curves at 0.95 without re-simulating anything, writing `minimum_texts_095.csv`. At that target the requirements are 926 distinct texts for lexical difficulty (95 per cent interval 794 to 1,069), 812 for grammatical load (723 to 906), 784 for referential continuity (696 to 877), 1,510 for the grammatical-load interaction with the age band (1,349 to 1,680), 1,506 for the narration interaction (1,349 to 1,672) and 681 children in tier 2 for the within-child term (543 to 834). The continuity interaction did not reach 0.95 within the 1,600 texts simulated.

The ordinal outcome behind Hypothesis 5 is handled by a separate script, `design_analysis/run_ordinal_analysis.R`, because a cumulative logit cannot be fitted by the routine the timing models use and treating a three-category rating as continuous would answer a different question. The script keeps the same design specification and the same seed discipline, generates the rating from it, retains each reading with the 30 per cent availability the structural sample showed, applies the protocol's five-rater floor inside every replicate, and fits a cumulative-logit counterpart of the registered model with `ordinal::clmm`, carrying the three components, the age band and a text intercept. The region of practical equivalence is applied symmetrically on the log-odds scale as a half-width of log(1.05), which is the region the main text registers. The ordinal sweep used 60 replicates per grid point, and fits whose standard errors were unusable were discarded, leaving 46, 52, 56 and 59 fits at 200, 400, 800 and 1,600 texts.

The result is the most demanding requirement in the design. At 200, 400, 800 and 1,600 generated texts, with twelve readings each, the 95 per cent interval on a coefficient whose true value is 0.10 log odds has a mean width of 0.44, 0.30, 0.21 and 0.15, and the probability of a determinate verdict never rises above 0.29. The target of 0.90 is therefore not reached within the range simulated, and no crossing is quoted from it. What the grid does establish is the scaling: the fitted exponents of interval width on the number of texts are

between -0.51 and -0.53 against a theoretical -0.5, and on that law the requirement projects to roughly 8,000 to 8,700 generated texts, which is about 2,300 to 2,500 texts carrying at least five ratings. Set against the 738 texts the same coefficient needs on reading time, the rating demands an order of magnitude more, which is what a three-category response observed for 30 per cent of readings costs. Both quantities depend on the rating's availability across the full window and on how heavily texts are reused, which the provider-computed counts will supply. The simulation will be re-run at the observed values before any outcome model is fitted. `design_analysis/summarise_ordinal.R` pools the cached grid points and reproduces every number in this paragraph.

S6. Deriving the priors from the systematic review

Priors for the confirmatory main effects and for the secondary text predictors come from the systematic review accompanying this project, through the pipeline whose freeze-and-run discipline is stated under Priors in the main text. This section records what the main text does not: the mapping from the review's extracted effects to this protocol's coefficients, the conversion order, and the pooling and de-duplication rules. The mapping is one-to-many and is fixed as follows. LexicalDifficulty pools the review's word-frequency, lexical-coverage, other-lexical and lexical-diversity effects on reading, sign-harmonised so that frequency and coverage are flipped to a difficulty direction while age of acquisition, word length and surprisal keep their sign. GrammaticalLoad pools syntactic-complexity and sentence-length effects. ReferentialContinuity pools coherence effects, with discourse-structure rows informing the secondary narrativity predictor instead. Valence and arousal pool the text-emotionality rows separately where the extraction distinguishes them.

Estimates will be put on a common scale in a fixed order of preference. A standardised beta or a correlation goes through Fisher's *z*. A *d* or *g* converts to *r* using the effective sample size, and an odds ratio converts to *d* and then to *r*. Where no effect size is reported but a test statistic is, *r* is recovered from *t* and its degrees of freedom and signed by the reported direction. Unstandardised coefficients without convertible scale information will not be pooled and will be listed in an exclusion appendix, as will rows whose direction the extraction could not resolve. Pooling will be a random-effects meta-analysis within each combination of coefficient and outcome class, with one estimate per paper per relation, so that labelled subgroups of a single paper are never treated as independent. Where a meta-analysis in the corpus subsumes three or more of the corpus's own primary studies, the pooled row will be preferred and the overlapping primaries dropped. Otherwise the primaries will be used and the pooled row dropped, and the alternative will be reported as a sensitivity analysis.

The prior is then Student's $t(3, \hat{\mu}, \sigma)$ with $\hat{\mu}$ the pooled estimate on the model's scale and $\sigma = 2\sqrt{\mathrm{SE}(\hat{\mu})^2 + \hat{\tau}^2}$, floored at 0.10. The heavy tails let the data override a misplaced centre. The complete prior table, the forest plots, the conversion arithmetic and the list of excluded rows will accompany the Stage 2 report, and the pipeline's output is timestamped and deposited before the extract is transferred. The same output records, for each component, whether the pooled estimate on the model's scale falls below 0.10 standard deviations, which is the comparison the main text's precision thresholds depend on.

S7. Data-access certification and activation record

We sent the text below to the data controller on 7 August 2026, and it is reproduced as it stands. The bracketed fields will be completed and the document signed before this protocol is submitted. [At submission: "The certification was signed on DATE and every field is completed above." / "The following fields remained open at submission: …".]

Gameloft confirms that, as of [date]:

1. An export of *Let's Story* reading sessions was built on [build date], covering [window] and comprising [n] rows, in the file or files listed in the accompanying record with their SHA-256 checksums.
2. That extract is held by Gameloft and has not been transferred, in whole or in part, to the University of Oxford research team or to anyone acting for it.
3. It will be transferred only on written confirmation of in-principle acceptance.
4. Material shared with the team to date: a 35-row sample and the data dictionary, shared on 27 July 2026 to confirm the structure of the table; the story texts; and the counts and file record described above.
5. No analysis relating story text to reading time has been conducted by Gameloft or reported to the team.

Owner: [name, role]
Deputy: [name, role]
Signed: [name, role, date]

Clause 4 needs three glosses. The 35-row sample carries real reading times, and the main text discloses it directly and reports it in Section S8. The summary of that sample, which we computed with the count script on 6 August 2026, is covered by the sample deposit and is listed under Pilot data in the main text. The story texts are named here because the editorial office confirmed in advance that stimulus materials may be received before submission, for the reasons given under Data provenance in the main text.

The accompanying record referred to in clause 1 is the manifest described in the main text, deposited with this protocol. Any row removed from the held extract between the build and the transfer is recorded by the data controller, as the Ethics statement sets out, and its effect on the counts will be reported at Stage 2.

This section is also where any written confirmation activating a contingent hypothesis would be reproduced. At submission, none of the activation conditions had been confirmed in writing by the data controller: item-level comprehension responses with a recoverable question type for C1; item-level responses together with accounts observed with and without narration, the latter read from the provider-computed counts, for C2 and C3; and fixed human-authored classic texts for C4. No contingent hypothesis is therefore activated. Should a condition be confirmed after submission, the analysis written for it would be reported at Stage 2 under Exploratory analyses, and a hypothesis whose condition is never confirmed is not run.

S8. The structural sample

The pilot material for this protocol is the 35-row structural sample and the data dictionary that the data controller shared on 27 July 2026 to confirm the shape of the table, and the summary of that sample computed with the count script described under Data provenance in the main text. The sample is what fixed three inputs of the design analysis: the narration prevalence, the rating availability and the age-band missingness. It cannot support any hypothesis, because it holds one session per account and five distinct texts.

The facts the sample established are these. Every one of the 35 sessions was a finished reading, with the last page reached equal to the page count, so the extract is a record of completed readings. Narration was on in 32 of the 35 sessions. The post-reading difficulty rating was present in 10 sessions and absent in 25. The age band was blank in 10 sessions, and the recorded bands skewed towards children under 7, which is why the protocol's inclusion criterion of 7 and older is expected to remove a substantial share of the export. The sessions covered ten reading languages. The per-session activity counters summed to more than the whole

reading duration in 6 of the 14 sessions with any activity, in one case 438 seconds against 222, which is why those fields are set aside. Reading times were recorded as whole seconds.

Supplementary Table 4 tabulates the 35 rows by field, and Supplementary Table 5 reproduces the reading-time check from the summary that the count script wrote on 6 August 2026. The script is deposited with the protocol in `count_script/`, and re-running it on the sample reproduces that summary line for line, so both tables can be regenerated from the sample deposit. One further fact about the file bears on the transfer. The account identifier reached us in floating-point notation with six significant digits, which is how a spreadsheet writes a nineteen-digit integer. The 35 values are distinct at that precision, but a full export written the same way could not be relied on to distinguish accounts. The sample itself, which carries real reading times, is deposited for peer review under embargo until Stage 2 acceptance, at the location the Data availability statement names.

Supplementary Table 4
The 35-row structural sample by field. Counts are of sessions unless stated.

Field	Rows with a value	Distinct values	Values, or note
user_id	35	35	One session per account. Six significant digits, as noted above.
game_version	35	10	6.2.0 to 6.7.2.
finished_reading_at	35	35	2 January to 30 June 2026, in UTC.
user_country	35	21	Germany, Italy and Poland 4 each; 18 further countries.
user_age_group	25	4	Age 3-4 (8), Age 5-6 (9), Age 7-8 (6), Age 9+ (2). Blank in 10.
story_title	35	2	Crafted story (20), classic story (15).
story_id	35	5	One classic text (15 sessions) and four attribute combinations (11, 7, 1 and 1 sessions).
story_location, story_hero, story_sidekick, story_plot, story_theme	20	1, 2, 2, 3, 3	Filled for crafted stories only.
is_customized	35	2	1 (20), 0 (15).
is_speech_on	35	2	1 (32), 0 (3).
time_spent_reading	35	32	Whole seconds. Checked through Supplementary Table 5 only, and no value is reported.
read_language	35	10	English 11, German 5, Polish 5, Italian 4, Spanish 3, Korean 2, Portuguese 2, Dutch 1, French 1, Greek 1.
user_reading_level	32	3	Age 3-5 (16), Age 5-7 (6), Age 7+ (10). Blank in 3.
max_page_reached, total_page_available	35	5	6, 7, 10, 11 or 17 pages. Equal to each other in every row.
story_rating	10	3	Just Right (7), Easy (2), Difficult (1). Blank in 25.
The eight minigame_*_ts fields	35	4 to 10	Zero throughout the 15 classic sessions and 6 crafted sessions. Non-zero in 14, of which 6 sum to more than the reading duration.
word_explorer_ts	35	8	Non-zero in 7 sessions. A lifetime total, by the dictionary.

Field	Rows with a value	Distinct values	Values, or note
inkmagination_ts	35	30	Non-zero in 29 sessions. A lifetime total, by the dictionary.

Supplementary Table 5

The reading-time check on the sample, as the count script reports it. The bands are ten times wider than one another and are never crossed with any other field.

Group	Sessions
Empty	0
Not a number	0
Negative	0
Zero	0
Above 0 but below 1	0
1 to 9	0
10 to 99	15
100 to 999	20
1,000 to 9,999	0
10,000 to 99,999	0
100,000 or more	0
Sessions with a usable number	35
Distinct values in the column	32
Sessions sharing the most common value	2

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